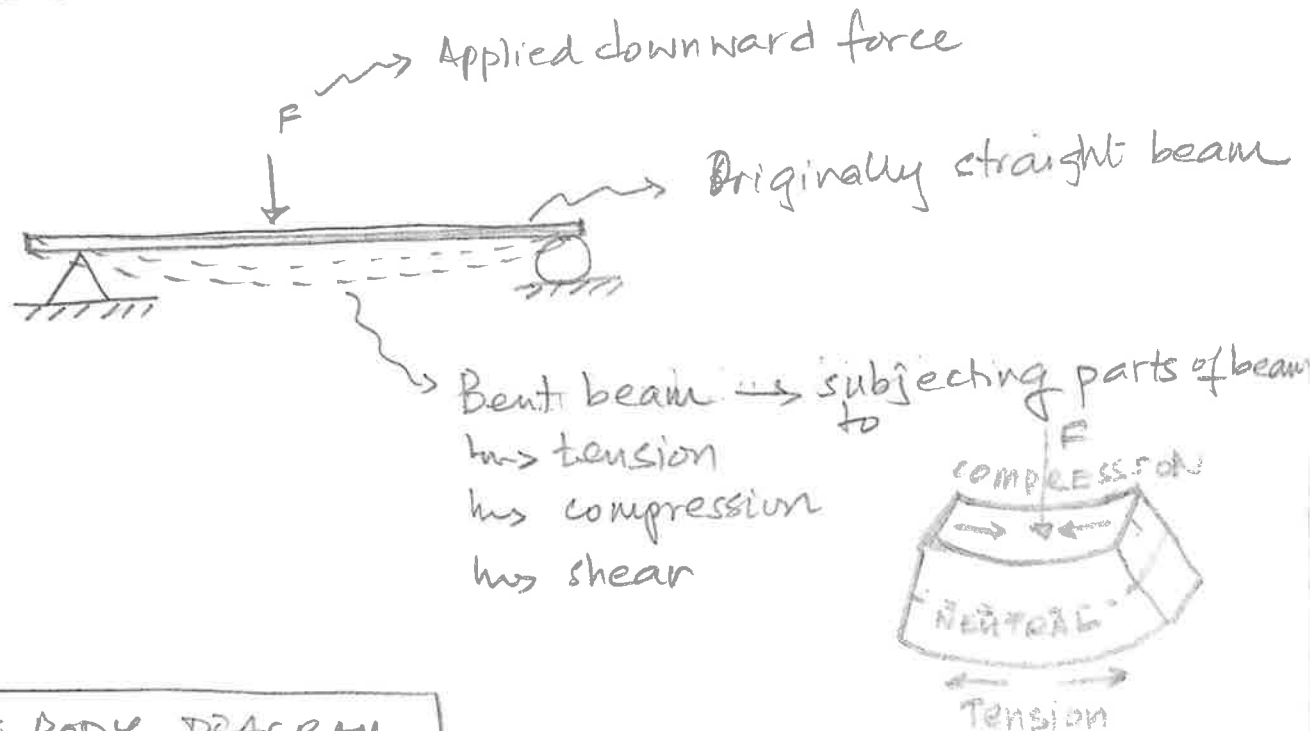
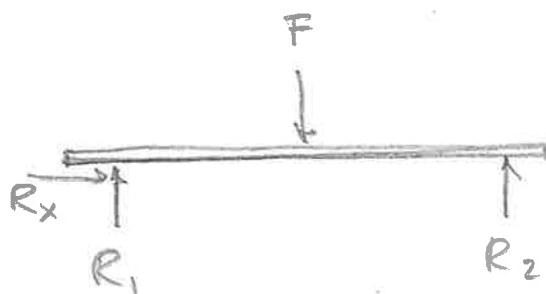


INTRODUCTION TO BENDING:

①

CONSIDER A SIMPLY SUPPORTED BEAMFREE BODY DIAGRAM

Three-point bending



Four-point bending



STRESS ANALYSIS
OF
STRUCTURES
SUBJECTED TO
BENDING

STATIC EQUILIBRIUM
ANALYSES

$\hookrightarrow$ To determine

- $\hookrightarrow$ Internal forces
- $\hookrightarrow$ External forces

DETERMINE STRESSES

②

EQUATIONS FOR 2D PROBLEMS: (X-Y coordinates)

$$\left. \begin{aligned} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned} \right\} \text{TRANSLATIONAL EQUILIBRIUM EQUATIONS}$$

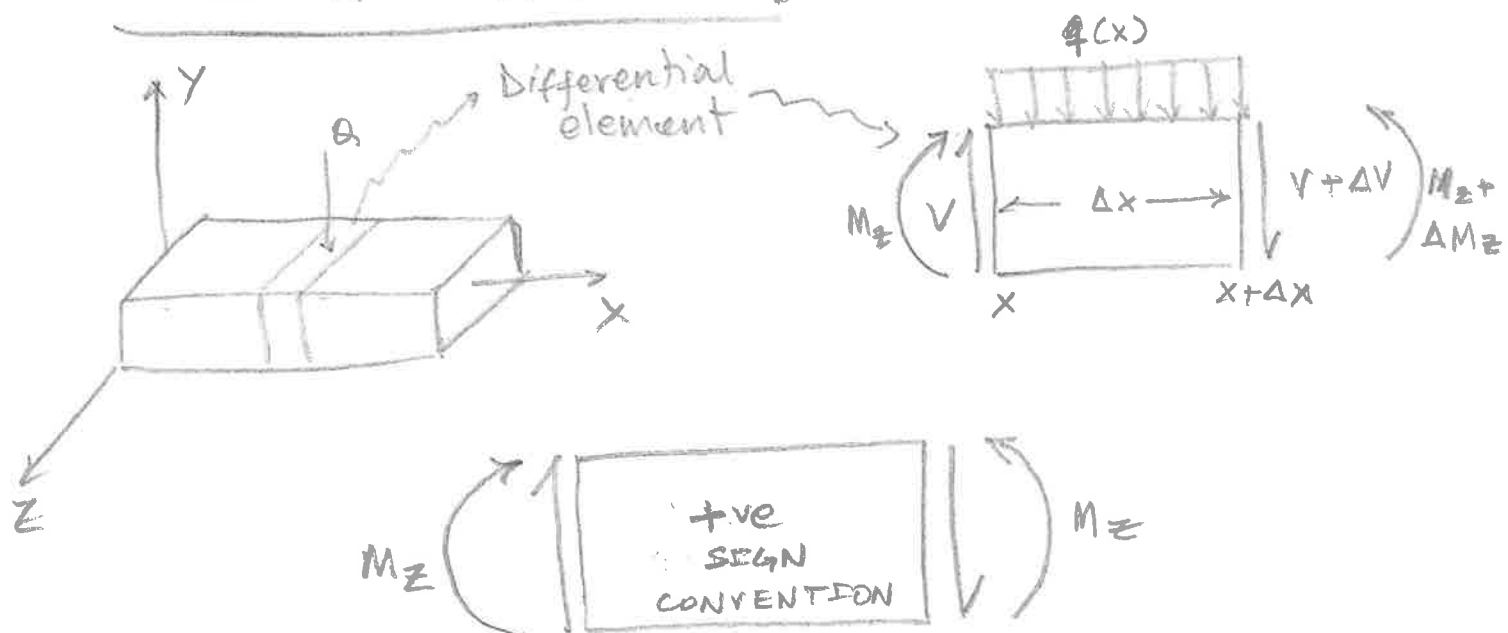
$$\sum M_z = 0 \quad \left. \right\} \text{ROTATION EQUILIBRIUM EQUATION}$$

METHOD OF SECTIONS: $\rightarrow$ To determine

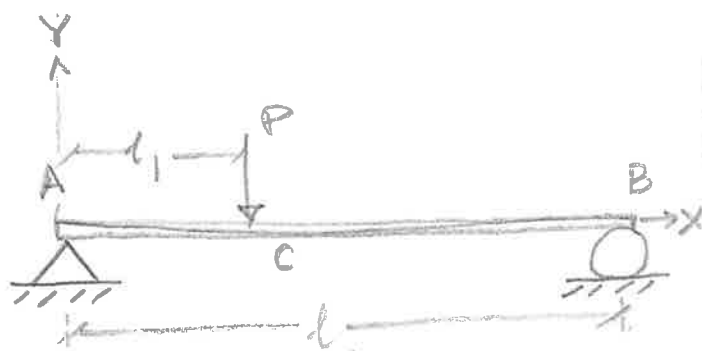
- $\rightarrow$ internal resistance of structures
- $\rightarrow$ due to externally applied forces

[Individual parts of structures that are in equilibrium should be in equilibrium]

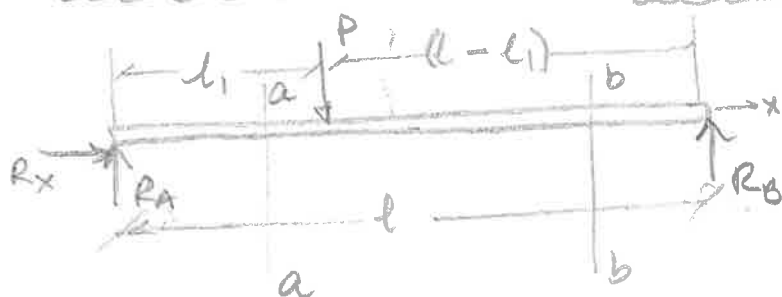
SIGN CONVENTION, EQUILIBRIUM ON A DIFFERENTIAL ELEMENT OF A STRUCTURE:



LET'S CONSIDER TWO EXAMPLES:



FREE BODY DIAGRAM OF THE FULL STRUCTURE



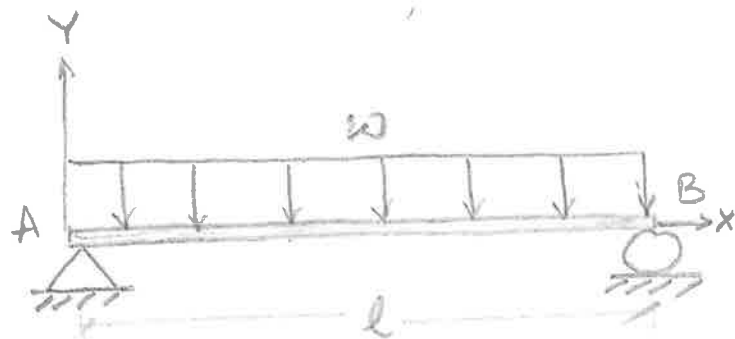
$$\sum F_x = 0 \Rightarrow R_x = 0$$

$$\sum F_y = 0 \Rightarrow R_A + R_B - P = 0$$

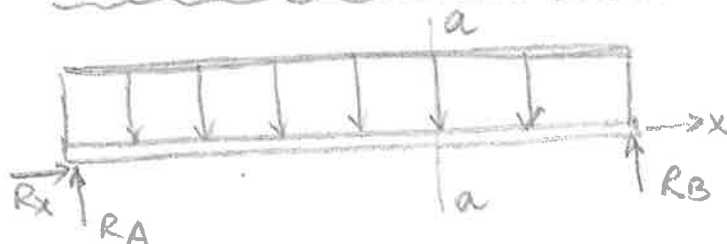
$$\sum M_A = 0 \Rightarrow R_B(l) - P(l_1) = 0$$

$$\Rightarrow R_B = \frac{P \cdot l_1}{l}$$

$$\Rightarrow R_A = P \left(1 - \frac{l_1}{l}\right)$$



FREE BODY DIAGRAM - FULL STRUCTURE



$$\sum F_x = 0 \Rightarrow R_x = 0$$

$$\sum F_y = 0 \Rightarrow R_A + R_B - \int_0^l w dx = 0$$

$$\Rightarrow R_A + R_B = wl$$

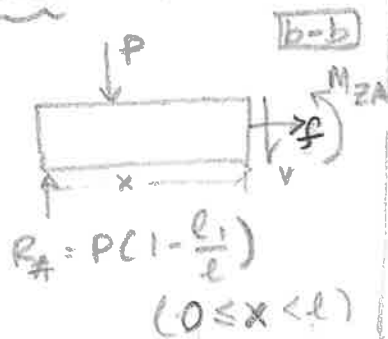
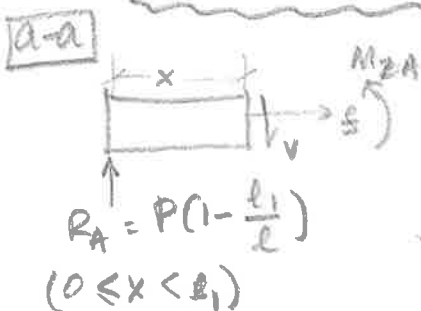
Assume load (wl) is acting at $\frac{l}{2}$ (centroid)

$$\sum M_A = 0 \Rightarrow R_B l - wl \left(\frac{l}{2}\right) = 0$$

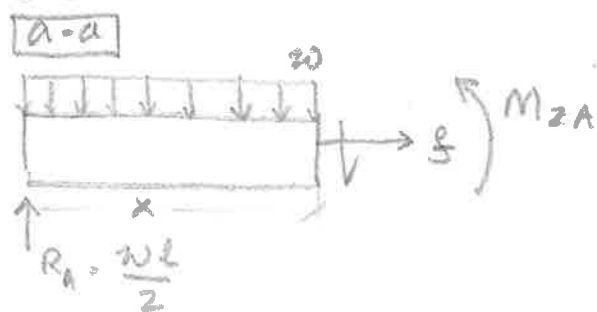
$$\Rightarrow R_B = \frac{wl}{2} \Rightarrow R_A = wl - \frac{wl}{2}$$

$$\Rightarrow R_A = \frac{wl}{2}$$

FREE BODY DIAGRAMS - SECTIONS

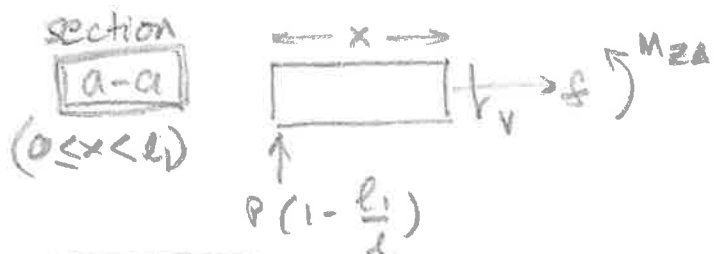


FREE BODY DIAGRAMS - SECTIONS



→ upto l, but not l

(#) WHERE V → SHEAR FORCE, f → FORCE along X-direction



$$\sum F_x = 0 \Rightarrow f = 0$$

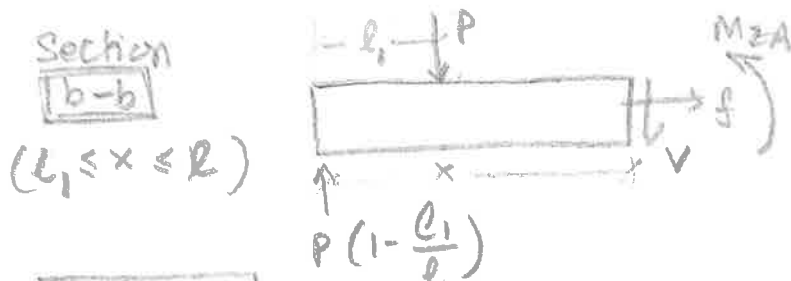
$$\sum F_y = 0 \Rightarrow P(1 - \frac{l_1}{l}) - V = 0$$

$$\Rightarrow V = P(1 - \frac{l_1}{l})$$

$$\sum M_{ZA} = 0 \Rightarrow M_{ZA} - (V)(x) = 0$$

(at section a-a) $\Rightarrow M_{ZA} = (V)(x)$

$$\Rightarrow M_{ZA}(x) = P(1 - \frac{l_1}{l})x$$



$$\sum F_x = 0 \Rightarrow f = 0$$

$$\sum F_y = 0 \Rightarrow P(1 - \frac{l_1}{l}) - P - V = 0$$

$$\Rightarrow V = P - P - P \frac{l_1}{l}$$

$$\Rightarrow V = -P \frac{l_1}{l}$$

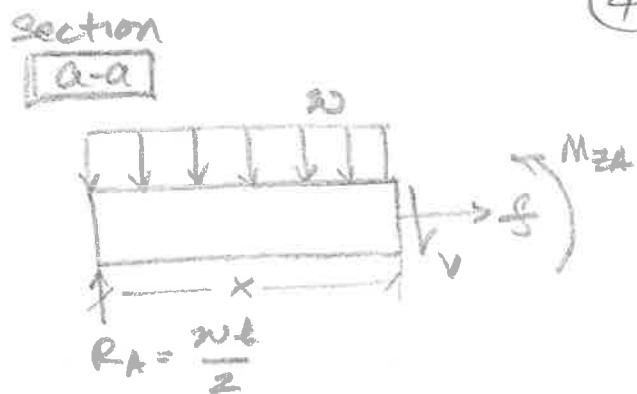
$$\sum M_{ZA} = 0 \Rightarrow M_{ZA} - Pl_1 - (V)(x) = 0$$

$$\Rightarrow M_{ZA} = Pl_1 + (V)(x)$$

$$= Pl_1 + (-\frac{Pl_1}{l})(x)$$

$$\Rightarrow M_{ZA}(x) = Pl_1(1 - \frac{x}{l})$$

(at section b-b)



$$\sum F_x = 0 \Rightarrow f = 0$$

$$\sum F_y = 0 \Rightarrow \frac{wl}{2} - wx - V = 0$$

$$\Rightarrow V = \frac{wl}{2} - wx$$

$$\sum M_{ZA} = 0 \Rightarrow M_{ZA} - (V)(x) - wx(\frac{x}{2}) = 0$$

Load (wx) is acting at $\frac{x}{2}$

$$\Rightarrow M_{ZA} = (V)(x) + (wx)(\frac{x}{2})$$

$$= (\frac{wl}{2} - wx)(x)$$

$$+ (wx)(\frac{x}{2})$$

$$= \frac{wlx}{2} - wx^2 + \frac{wx^2}{2}$$

$$\Rightarrow M_{ZA}(x) = \frac{wlx}{2} - \frac{wx^2}{2}$$

(at section a-a)

This is a quadratic equation ④

Maximum bending moment occurs when

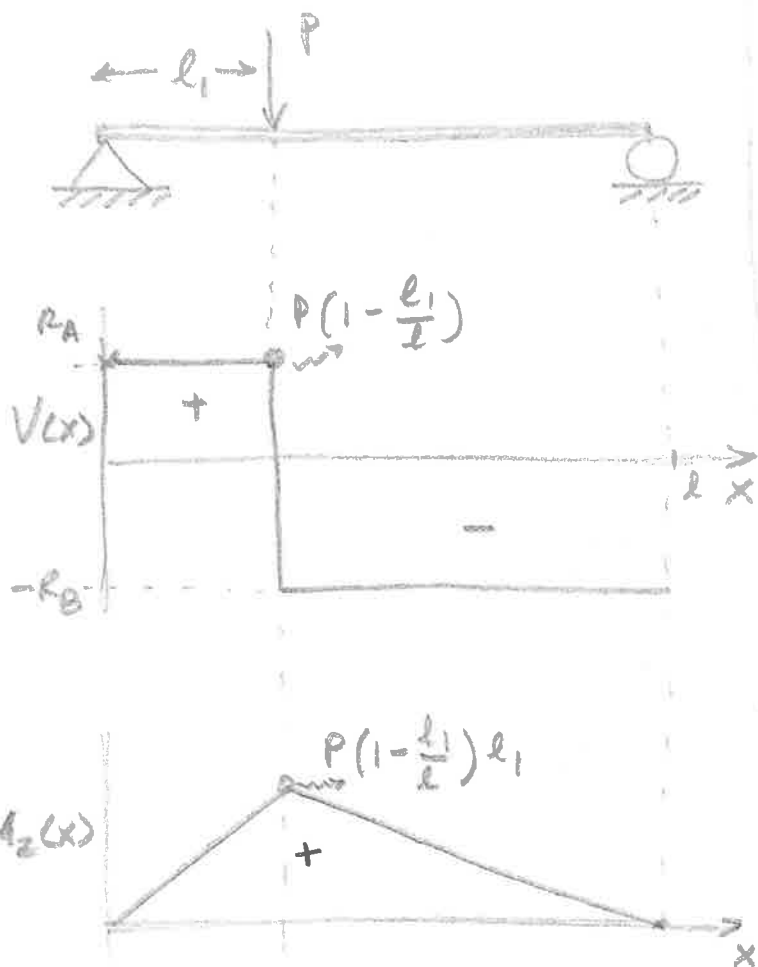
$$\frac{dM_{ZA}(x)}{dx} = 0$$

Shear force results:

$$V(x) = \begin{cases} P(1 - \frac{l_1}{l}) & 0 \leq x < l_1 \\ -P(\frac{l_1}{l}) & l_1 \leq x < l \end{cases}$$

Bending moment results:

$$M(x) = \begin{cases} P(1 - \frac{l_1}{l})x & 0 \leq x < l_1 \\ P l_1 (1 - \frac{x}{l}) & l_1 \leq x < l \end{cases}$$



Maximum bending moment location

$$\frac{dM_{2x}(x)}{dx} = \frac{d}{dx} \left[\frac{wlx}{2} - \frac{wx^2}{2} \right] = 0$$

$$\Rightarrow \frac{wl}{2} - \frac{2wx}{2} = 0$$

$$\Rightarrow \frac{w}{2} (l - x) = 0 \Rightarrow \boxed{x = \frac{l}{2}}$$

② This is the location along \$x\$-axis where the bending moment is maximum.

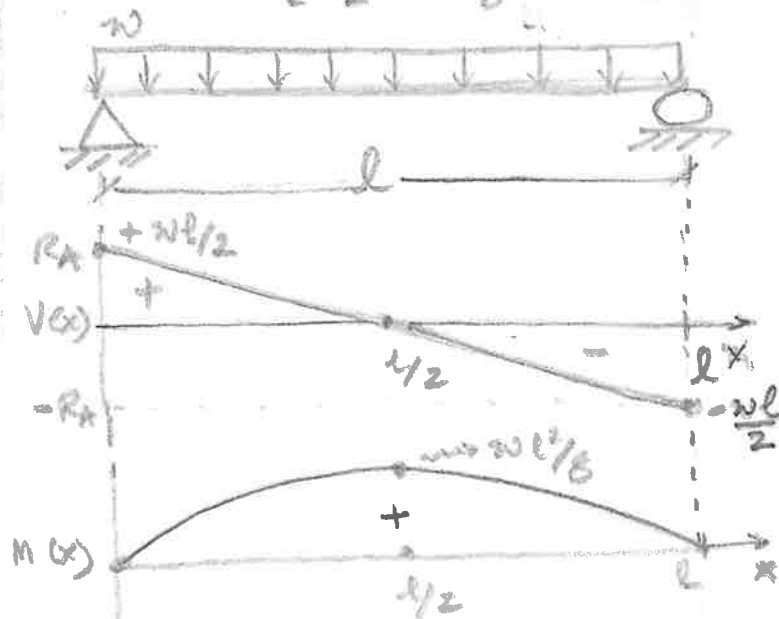
Shear force results:

$$V(x) = \frac{wl}{2} - wx$$

Bending moment results:

$$M(x) = \frac{wlx}{2} - \frac{wx^2}{2}$$

$$M(l/2) = \frac{wl}{2} \frac{l}{2} - \frac{wl^2}{8} = \frac{wl^2}{8}$$

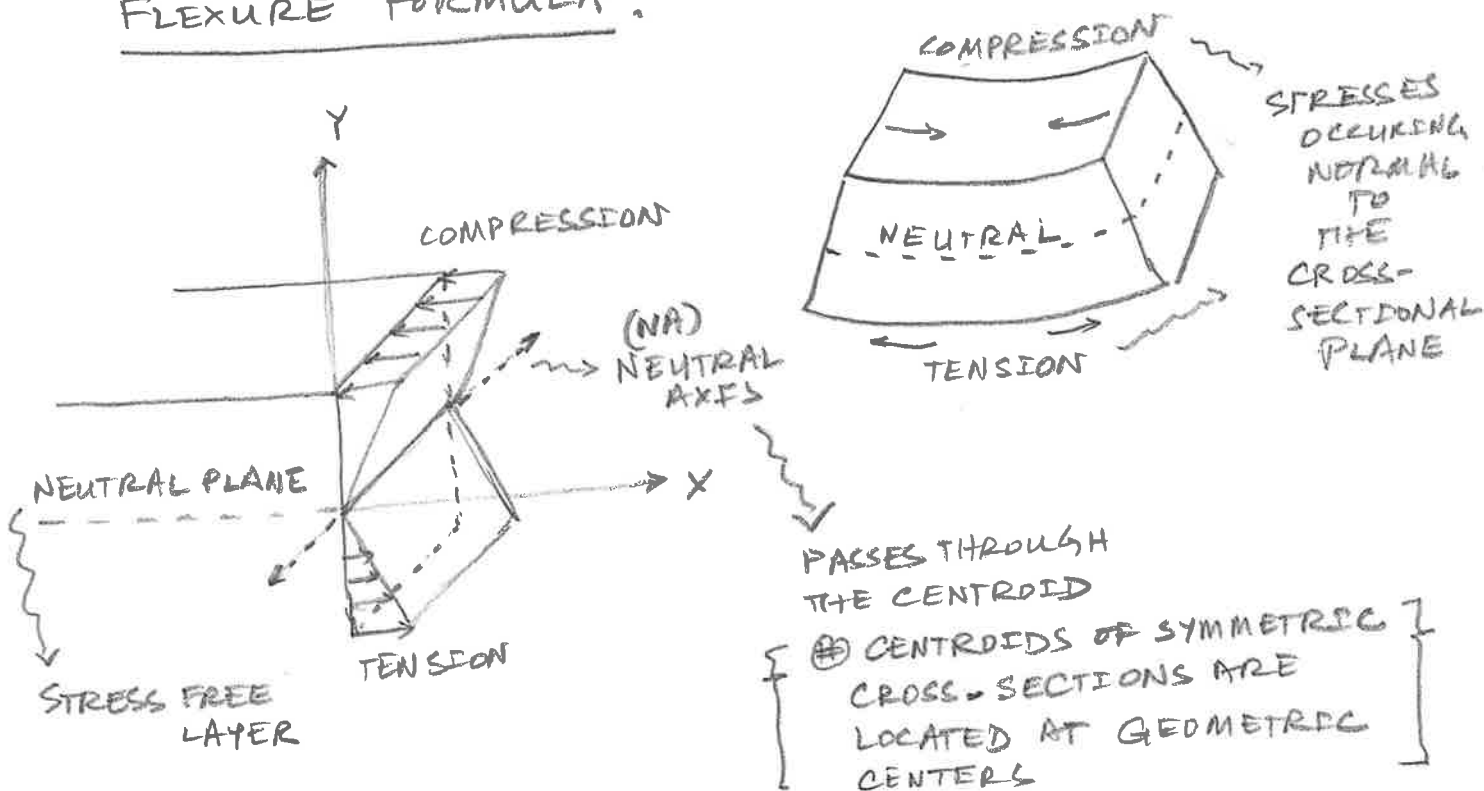


SHEAR FORCE & BENDING MOMENT RELATIONSHIP:

shear force $\rightarrow V = - \frac{dM}{dx}$ $\rightarrow$ Bending moment

(#) If variation of M along x -direction is known, shear force at any given section of the structure can be determined.

FLEXURE FORMULA: (#) BEAM IS SUBJECTED TO PURE BENDING



From equilibrium conditions: $\rightarrow$ Bending moment

Flexural stress $\rightarrow \sigma_x = - \frac{My}{I}$ $\rightarrow$ vertical distance from neutral axis where stress is sought.

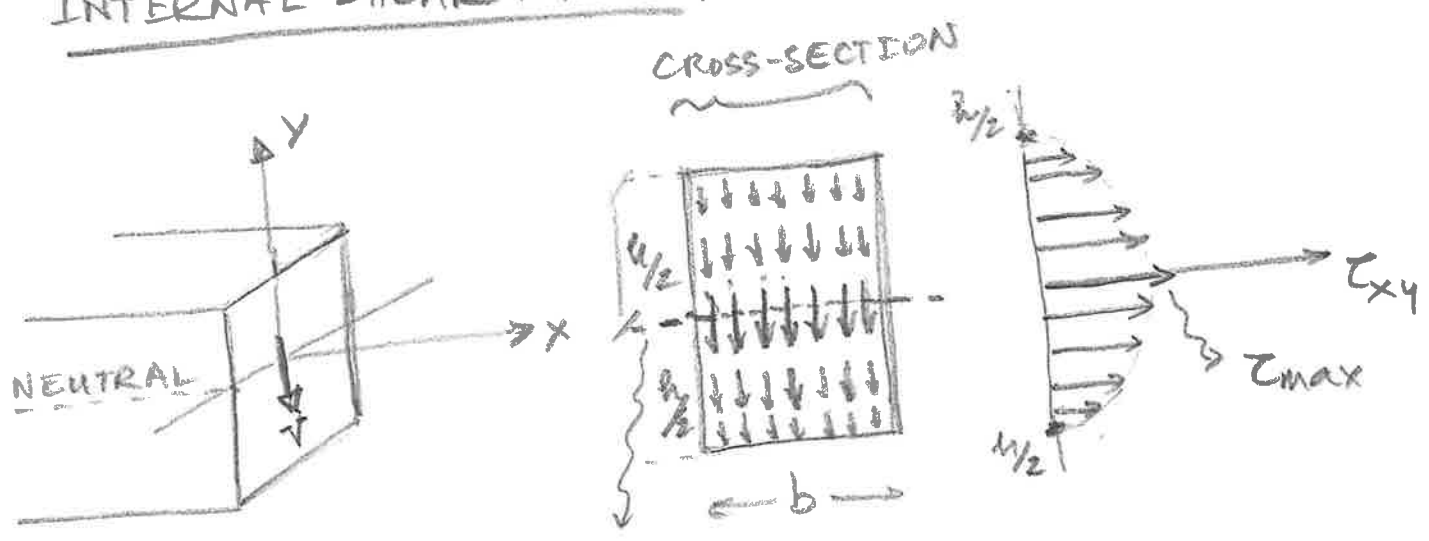
$\rightarrow$ Area moment of inertia of beam cross-section

Flexural stress $\left\{ \begin{array}{l} \sigma_x = 0 \text{ at neutral axis } y = 0 \\ \sigma_x \text{ is negative for } y > 0 : \text{compression} \\ \sigma_x \text{ is positive for } y < 0 : \text{tension} \end{array} \right.$

$\sigma_{\max} = \frac{M C}{I}$ $\rightarrow$ maximum value of y

$\underbrace{\hspace{1cm}}$
Magnitude
{ $\therefore$ -ve sign? }
{ is dropped }

INTERNAL SHEAR FORCE :



Neutral (NA) axis

$\tau_{xy} = \frac{V Q}{I b}$

$\left\{ \begin{array}{l} \text{Shear force at the section?} \\ \text{where } \tau_{xy} \text{ is sought} \end{array} \right.$

$\left\{ \begin{array}{l} \text{first moment of area about?} \\ \text{neutral axis} \end{array} \right.$

$\left\{ \begin{array}{l} \text{width of cross-section} \end{array} \right.$

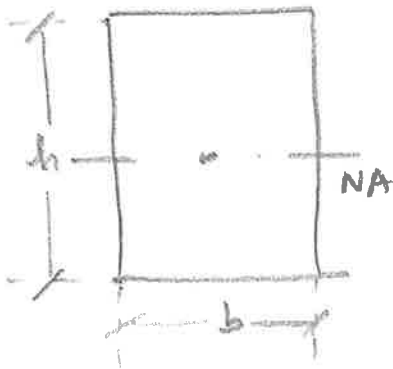
$\left\{ \begin{array}{l} \text{Shear stress} \\ \text{due to} \\ \text{bending} \end{array} \right.$

$\left\{ \begin{array}{l} \text{Moment} \\ \text{of inertia of} \\ \text{cross-sectional} \\ \text{area about neutral} \\ \text{axis} \end{array} \right.$

(8)

$$\sigma_x = -\frac{My}{I}$$

$$Z_{xy} = \frac{VQ}{Ib}$$



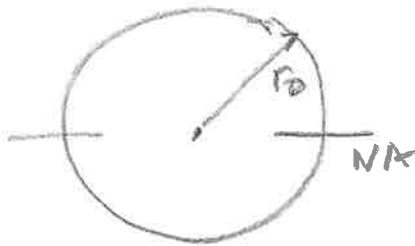
$$A = bh$$

$$I = \frac{bh^3}{12}$$

$$Q = \frac{bh^2}{8}$$

$$\sigma_{max} = \frac{Mh}{2I}$$

$$\tau_{max} = \frac{3V}{2A}$$



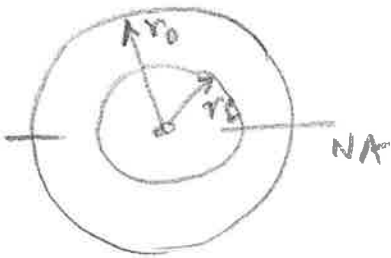
$$A = \pi r_0^2$$

$$I = \frac{\pi r_0^4}{4}$$

$$Q = \frac{2r_0^3}{3}$$

$$\sigma_{max} = \frac{Mr_0}{I}$$

$$\tau_{max} = \frac{4V}{3A}$$



$$A = \pi(r_0^2 - r_i^2)$$

$$I = \frac{\pi(r_0^4 - r_i^4)}{4}$$

$$Q = \frac{2}{3}(r_0^2 + r_i^2)$$

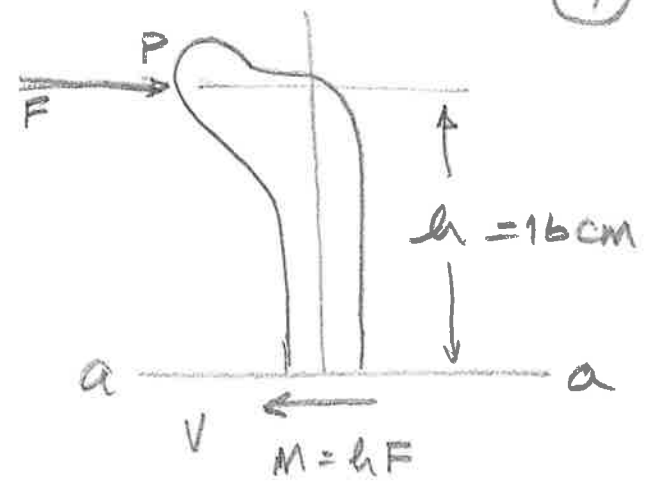
$$\sigma_{max} = \frac{Mr_0}{I}$$

$$\tau_{max} = \frac{4V}{3A} \left(\frac{r_0^2 + r_0 r_i + r_i^2}{r_0^2 + r_i^2} \right)$$

PROBLEM

The distal end of a human femur is clamped to a bench and a horizontal force with magnitude

$F = 500 \text{ N}$ is applied to the head as shown at point P.



Determine:

- (1) Maximum normal and shear stresses generated at section a-a,

Given that the femur has an inner radius $r_i = 6 \text{ mm}$ and outer radius $r_o = 13 \text{ mm}$ at section a-a.

SOLUTION

$$V = F = 500 \text{ N}$$

$$M = hF = (0.16) \text{ m} (500 \text{ N}) = 80 \text{ Nm}$$

$$A = \pi(r_o^2 - r_i^2) = 4.18 \times 10^{-4} \text{ m}^2$$

$$I = \frac{\pi}{4}(r_o^4 - r_i^4) = 2.14 \times 10^{-8} \text{ m}^4$$

$$Q = \frac{2}{3}(r_o^3 - r_i^3) = 1.32 \times 10^{-6} \text{ m}^3$$

$$\begin{aligned} \tau_{\max} &= \frac{Mr_o}{I} = \frac{(80 \text{ Nm})(0.013 \text{ m})}{2.14 \times 10^{-8} \text{ m}^4} \\ &= 48.6 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \tau_{\max} &= \frac{4V}{3A} \left(\frac{r_o^2 + r_or_i + r_i^2}{r_o^2 + r_i^2} \right) \\ &= \frac{4(500 \text{ N})}{3(4.18 \times 10^{-4} \text{ m}^2)} \left(\right) \\ &= 2.2 \text{ MPa} \end{aligned}$$